OMNIX
Decision Governance Infrastructure

POST-QUANTUM MANDATE RECEIPT
Schema Specification — Velos Integration Document
Schema v6.5.4e | ADR-022, ADR-031, ADR-043, ADR-044 | T=0 Gateway Edition

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Classification Confidential — Partner Integration Material

Schema Version v6.5.4e

Engine Version 6.5.4e

Signing Standard NIST FIPS 204 — ML-DSA (Dilithium-3 / Dilithium-5)

KEM Standard NIST FIPS 203 — ML-KEM (Kyber-768)

Purpose T=0 Interceptor Gateway — Velos POST endpoint integration

Every decision emitted by OMNIX produces a Post-Quantum Decision Receipt — a cryptographically signed JSON object encoding the complete governance trail of a single enforcement decision. This document defines the exact schema Velos must implement to correctly verify receipts at its T=0 interceptor gateway.

OMNIX Decision Governance Infrastructure — Post-Quantum Schema v6.5.4e — Confidential / Velos Integration — Page 1 of 12

1. Overview

The OMNIX receipt is the atomic unit of the evidence layer. Each receipt is:
• Signed using NIST FIPS 204 — Dilithium-3 (ML-DSA-65) at enterprise baseline, or Dilithium-5 (ML-DSA-87) at
high-assurance tier
• Chained — each receipt includes the SHA-256 hash of its predecessor, forming a tamper-evident audit trail
• Time-bounded — carries explicit ttl_epoch_ms for T=0 gateway enforcement
• Domain-agnostic — identical schema for trading, Islamic credit, insurance, and robotics

ENDPOINT: POST https://velos-gateway.onrender.com/api/v1/intercept All OMNIX receipts sent to this endpoint must pass TTL check + hash integrity + PQC signature verification before any downstream action is triggered.

2. Complete JSON Field Reference

{
"receipt_id": "OMNIX-A3F7C1D92B0E",
"timestamp": "2026-04-09T14:23:01.847291+00:00",
"issued_at_ms": 1744208581847,
"ttl_epoch_ms": 1744208611847,
"ttl_ms": 30000,

"asset": "BTC/USDT",
"decision": "BLOCK",

"veto_chain": [
"AML: PASS — volume_score=12.4 frequency_score=0.3", "FRAUD: PASS — sentiment_score=0.12 reversal_risk=0.04", "SHARIA: BLOCK — gharar_score=0.81 exceeds threshold 0.35", "CAG: SKIPPED (upstream veto)"
],

"policy_version": "6.5.4e",
"engine_version": "6.5.4e",
"prev_hash": "a9f3c1d4e8b2fc71...",
"signing_provider": "dilithium3",
"content_hash": "3f8e7a2c91bd...",

"signature": "BASE64_PQC_SIGNATURE==", "signature_algorithm": "Dilithium-3 (ML-DSA-65)", "signature_format": "base64_pqc",
"public_key": "BASE64_PUBLIC_KEY==",

// Conditional — only present if gate was evaluated: "sharia_compliance": { ... },
"aml_compliance": { ... },

OMNIX Decision Governance Infrastructure — Post-Quantum Schema v6.5.4e — Confidential / Velos Integration — Page 2 of 12

"fraud_compliance": { ... },

"jurisdiction_compliance": { ... },

"context_admission": { ... },

"veto_type": "SHARIA_BLOCK",

"avm_result": { ... }

}

3. Identity Fields (always present)

Field Type Description

receipt_id string Globally unique receipt identifier. Format: OMNIX-{12 hex chars uppercase}. Example:
OMNIX-A3F7C1D92B0E

timestamp string ISO-8601 UTC timestamp of receipt generation. Example:
2026-04-09T14:23:01.847291+00:00

4. Timing Fields — T=0 Gateway Critical (always present from v6.5.4e)

These three fields are the primary mechanism for Velos T=0 enforcement. They are part of the signed payload (included in content_hash). A receipt is expired when: now_ms > ttl_epoch_ms

Field Type Description

issued_at_ms integer Unix epoch milliseconds when the receipt was generated. Derived from timestamp.
This is the canonical T=0 reference point for the decision.

ttl_epoch_ms integer Unix epoch milliseconds when the receipt expires. ttl_epoch_ms = issued_at_ms +
ttl_ms. Velos MUST reject any receipt where now_ms > ttl_epoch_ms.

ttl_ms integer Time-to-live in milliseconds. Default: 30000 (30 seconds). Configurable on OMNIX side
via OMNIX_RECEIPT_TTL_MS environment variable.

Velos T=0 ingestion rule:

now_ms = int(time.time() * 1000)

if now_ms > receipt['ttl_epoch_ms']:

return {'status': 'REJECTED', 'reason': 'RECEIPT_EXPIRED'}

5. Decision Fields (always present)

Field Type Description

asset string Asset symbol. Examples: BTC/USDT, ETH/USD, UNKNOWN

decision string Final governance decision, uppercased. Values: BLOCK | ALLOW | PASS |
UNKNOWN. BLOCK means OMNIX has vetoed the action — Velos must not proceed.

veto_chain array[string] Ordered list of gate verdicts. Each entry format: 'GATE_NAME: RESULT — detail'.
Maximum 80 characters per entry. Empty array if no trace available.

OMNIX Decision Governance Infrastructure — Post-Quantum Schema v6.5.4e — Confidential / Velos Integration — Page 3 of 12

6. Versioning Fields (always present)

Field Type Description

policy_version string Policy version that evaluated this decision. Usually matches engine_version.

engine_version string OMNIX engine version. Current production version: 6.5.4e

prev_hash string SHA-256 content_hash of the immediately preceding receipt in the chain. Empty string
for the first receipt. Forms the tamper-evident chain.

OMNIX Decision Governance Infrastructure — Post-Quantum Schema v6.5.4e — Confidential / Velos Integration — Page 4 of 12

7. Cryptographic Fields (always present)

Field Type Description

signing_provider string Identifier of the active signing algorithm. Values: dilithium3 | dilithium5 | ed25519 |
sha256. Bound into the signed payload — prevents algorithm confusion and
downgrade attacks. Velos MUST dispatch verification based on this field.

content_hash string SHA-256 hex digest of the canonical JSON of all mandatory + present conditional
fields, serialized with sort_keys=True, ensure_ascii=True. This is what gets signed.

signature string|null Detached digital signature over content_hash.encode('utf-8'). Encoding depends on
signature_format — see Section 8.

signature_algorithm string Human-readable algorithm name. Examples: Dilithium-3 (ML-DSA-65), SHA-256,
NONE.

signature_format string Critical for Velos ingestion. Disambiguates the encoding of the signature field. See
Section 8 for all possible values and how to handle each.

public_key string|null Base64-encoded public key for asymmetric signature verification. null in SHA-256
fallback mode (signature_format = hex_sha256_fallback).

8. signature_format — Dispatch Table

This field is the key switch for how Velos must process the signature. Never assume — always read signature_format first.

Value Meaning Verification

base64_pqc signature is Base64-encoded raw PQC signature
bytes.

base64.b64decode(signature) → verify with
Dilithium/ML-DSA using public_key

hex_sha256_fallback signature is hex of SHA-256(content_hash).
Symmetric. No asymmetric signing. public_key will
be null.

sha256(content_hash.encode()).hexdigest() ==
signature

NONE Signing failed. Receipt has no integrity guarantee. REJECT this receipt unconditionally.

9. Conditional Governance Blocks

These blocks appear only when the corresponding gate was evaluated for the decision. Structure varies by domain and gate version.

Field Present when

sharia_compliance Islamic finance gate was evaluated

aml_compliance AML gate was evaluated

fraud_compliance Fraud gate was evaluated

jurisdiction_compliance Jurisdiction gate was evaluated

context_admission CAG (Context Admission Gate) was evaluated

OMNIX Decision Governance Infrastructure — Post-Quantum Schema v6.5.4e — Confidential / Velos Integration — Page 5 of 12

veto_type context_admission is present and contains a veto_type string

avm_result AVM (Assumption Validity Monitor) was evaluated

avm_result sub-schema:
{
"is_valid": true,
"snapshot_id": "snap-abc123",
"parameter_version": "v3.1",
"drift_score": 0.12,
"age_hours": 2.4,
"pass_through": false,
"block_reason": "DRIFT_EXCEEDED" // only if AVM blocked }

OMNIX Decision Governance Infrastructure — Post-Quantum Schema v6.5.4e — Confidential / Velos Integration — Page 6 of 12

10. content_hash Contract — What Gets Hashed

The content_hash is computed before signing:

canonical = json.dumps(payload_fields, sort_keys=True, ensure_ascii=True) content_hash = sha256(canonical.encode('utf-8')).hexdigest()

Mandatory fields in hash:

asset
decision
engine_version
issued_at_ms (new v6.5.4e)
policy_version
prev_hash
receipt_id
signing_provider
timestamp
ttl_epoch_ms (new v6.5.4e)
ttl_ms (new v6.5.4e)
veto_chain

Conditional (if present):
aml_compliance
avm_result
context_admission
fraud_compliance
jurisdiction_compliance
sharia_compliance
veto_type

NOT included in hash:

content_hash (itself)
signature
signature_algorithm
signature_format
public_key

11. Full Verification Algorithm for Velos

Drop this function directly into the Velos interceptor. No external dependencies beyond pypqc (pip install pypqc).

import json, hashlib, base64, time

def verify_omnix_receipt(receipt: dict) -> dict:
result = {'receipt_id': receipt.get('receipt_id'), 'valid': False, 'reason': None}

# Step 1 — TTL check
ttl_epoch_ms = receipt.get('ttl_epoch_ms')
if ttl_epoch_ms is not None:
now_ms = int(time.time() * 1000)
if now_ms > ttl_epoch_ms:
result['reason'] = 'RECEIPT_EXPIRED'
return result

# Step 2 — Reconstruct payload for hashing
MANDATORY = ['receipt_id', 'timestamp', 'issued_at_ms', 'ttl_epoch_ms', 'ttl_ms', 'asset', 'decision', 'veto_chain',
'policy_version', 'engine_version', 'prev_hash', 'signing_provider']
CONDITIONAL = ['sharia_compliance', 'aml_compliance', 'fraud_compliance', 'jurisdiction_compliance', 'context_admission', 'veto_type', 'avm_result']

OMNIX Decision Governance Infrastructure — Post-Quantum Schema v6.5.4e — Confidential / Velos Integration — Page 7 of 12

payload = {k: receipt[k] for k in MANDATORY if k in receipt} if payload.get('signing_provider') is None: payload.pop('signing_provider', None)
for k in CONDITIONAL:
if k in receipt:
payload[k] = receipt[k]

# Step 3 — Verify content_hash
canonical = json.dumps(payload, sort_keys=True, ensure_ascii=True) computed_hash = hashlib.sha256(canonical.encode('utf-8')).hexdigest() if computed_hash != receipt.get('content_hash'):
result['reason'] = 'HASH_MISMATCH'
return result

# Step 4 — Verify signature based on signature_format
sig_format = receipt.get('signature_format', 'UNKNOWN')
signature = receipt.get('signature')
public_key = receipt.get('public_key')
message = receipt['content_hash'].encode('utf-8')

if sig_format == 'base64_pqc':
provider = receipt.get('signing_provider', 'dilithium3') if provider == 'dilithium3':
from pqc.sign import dilithium3
try:
dilithium3.verify(base64.b64decode(signature), message, base64.b64decode(public_key))
result['valid'] = True
except Exception:
result['reason'] = 'SIGNATURE_INVALID'
elif provider == 'dilithium5':
from pqc.sign import dilithium5
try:
dilithium5.verify(base64.b64decode(signature), message, base64.b64decode(public_key))
result['valid'] = True
except Exception:
result['reason'] = 'SIGNATURE_INVALID'

elif sig_format == 'hex_sha256_fallback':
expected = hashlib.sha256(receipt['content_hash'].encode()).hexdigest() result['valid'] = (expected == signature)
if not result['valid']:
result['reason'] = 'FALLBACK_HASH_MISMATCH'

OMNIX Decision Governance Infrastructure — Post-Quantum Schema v6.5.4e — Confidential / Velos Integration — Page 8 of 12

else:

result['reason'] = f'UNKNOWN_SIGNATURE_FORMAT:{sig_format}'

return result

OMNIX Decision Governance Infrastructure — Post-Quantum Schema v6.5.4e — Confidential / Velos Integration — Page 9 of 12

12. Cryptographic Assurance Tiers

Tier Provider ID Algorithm Standar
d

Security PK
Size

Sig Size

Enterprise Baseline
(default)

dilithium3 Dilithium-3 (ML-DSA-65) FIPS 204 ~192-bit 1952 B ~3309 B

High-Assurance dilithium5 Dilithium-5 (ML-DSA-87) FIPS 204 ~256-bit 2592 B ~4627 B

Dev/Test Fallback ed25519 Ed25519 (classical) — 128-bit 32 B 64 B

No PQC Available sha256 SHA-256 (symmetric
fallback)

— N/A N/A 64 hex

Tier is set on OMNIX side via PQC_SIGNING_LEVEL env var (3 or 5). Velos MUST dispatch verification based on the signing_provider field embedded in the receipt — do not hardcode a provider.

13. Receipt Chaining — Tamper-Evident Audit Trail

Each receipt carries prev_hash = content_hash of the immediately preceding receipt:

Receipt[n].prev_hash == Receipt[n-1].content_hash

A chain break indicates:

• A receipt was injected out of order

• A receipt was tampered with

• Chain was restarted — first receipt always has prev_hash = ""

14. Integration Checklist — Velos T=0 Gateway

Action Rule

1 Extract ttl_epoch_ms Reject if now_ms > ttl_epoch_ms

2 Extract signature_format Dispatch verification accordingly. DO NOT assume base64.

3 Extract signing_provider Determines which PQC algorithm to use for verification.

4 Install pypqc pip install pypqc on Velos side.

5 Verify content_hash Recompute from payload. Exclude: signature, public_key, signature_format,
signature_algorithm, content_hash itself.

6 Check decision field BLOCK = OMNIX has vetoed. Velos must not proceed.

7 Read veto_chain Identify which specific gate(s) triggered the block.

8 Reject signature_format
NONE

Treat as invalid receipt — no integrity guarantee.

OMNIX Decision Governance Infrastructure — Post-Quantum Schema v6.5.4e — Confidential / Velos Integration — Page 10 of 12

15. Example — Full BLOCK Receipt (Trading Domain)

{
"receipt_id": "OMNIX-A3F7C1D92B0E",
"timestamp": "2026-04-09T14:23:01.847291+00:00",
"issued_at_ms": 1744208581847,
"ttl_epoch_ms": 1744208611847,
"ttl_ms": 30000,
"asset": "BTC/USDT",
"decision": "BLOCK",
"veto_chain": [
"AML: PASS -- volume_score=12.4 frequency_score=0.3", "FRAUD: PASS -- sentiment_score=0.12 reversal_risk=0.04", "SHARIA: BLOCK -- gharar_score=0.81 exceeds threshold 0.35", "CAG: SKIPPED (upstream veto)"
],
"policy_version": "6.5.4e",
"engine_version": "6.5.4e",
"prev_hash": "a9f3c1d4e8b2fc71...",
"signing_provider": "dilithium3",
"sharia_compliance": {
"admitted": false,
"evaluation_state": "EVALUATED",
"violation": "GHARAR_EXCESSIVE",
"gharar_score": 0.81,
"debt_ratio": 0.0
},
"content_hash": "3f8e7a2c91bd...",
"signature": "BASE64_DILITHIUM3_SIGNATURE==", "signature_algorithm": "Dilithium-3 (ML-DSA-65)", "signature_format": "base64_pqc",
"public_key": "BASE64_PUBLIC_KEY=="
}

16. Backward Compatibility

Schema Version Released Changes

Schema v6.5.4e (this
document)

2026-04-0
9

Added issued_at_ms, ttl_epoch_ms, ttl_ms, signature_format. Verifier now checks TTL
expiry and exposes is_expired, age_ms.

Pre-v6.5.4e (legacy) — timestamp only. No TTL fields. signature_format absent. Assume base64_pqc if public_key present, else hex_sha256_fallback.

OMNIX Decision Governance Infrastructure — Post-Quantum Schema v6.5.4e — Confidential / Velos Integration — Page 11 of 12

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